EXPLAINABILITY OF VISION TRANSFORMERS

Inherently Explainable and Task-specific Methods, Evaluation, Tools and Outlook

Sections 3.4–8 of Kashefi et al., *Explainability of Vision Transformers: A Comprehensive Review and New Perspectives* (arXiv:2311.06786, 2023)

[Your name] [Teammates]

University of Science - VNUHCM

Demo 3.2: we re-run Fig. 3 of the survey

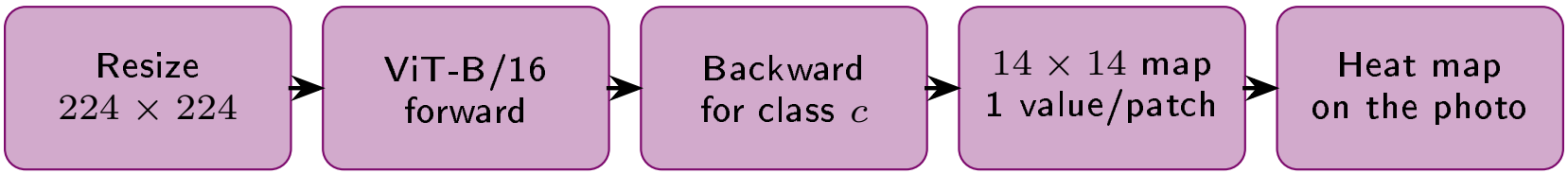


Dog 243 Cat 282



Elephant 101 Zebra 340

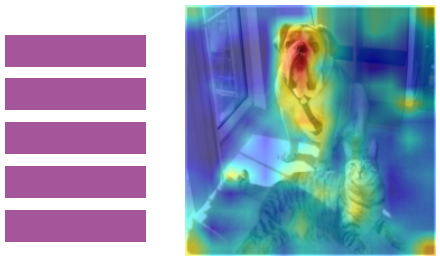
for each photo and class c



Model and code: the **authors' official implementation** (Chefer et al.)

Six columns, six ways to ask "which patches matter?"

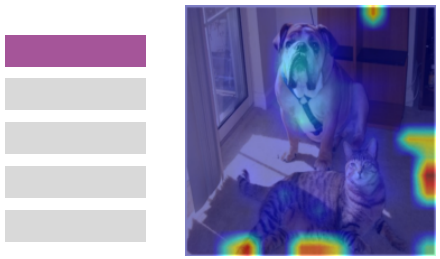
rollout



12 layers multiplied

same for any class

raw-attention



last layer only

same for any class

GradCAM



last layer × gradient

class-specific

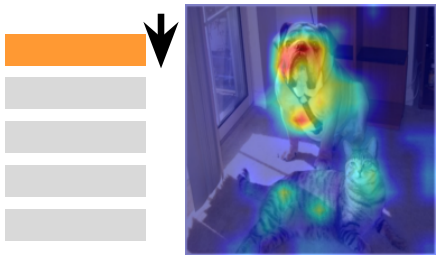
LRP



back to the pixels

weak

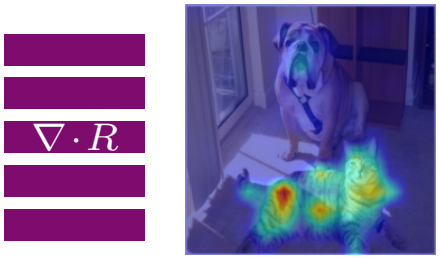
partial LRP



last layer only

partly

Beyond Attention

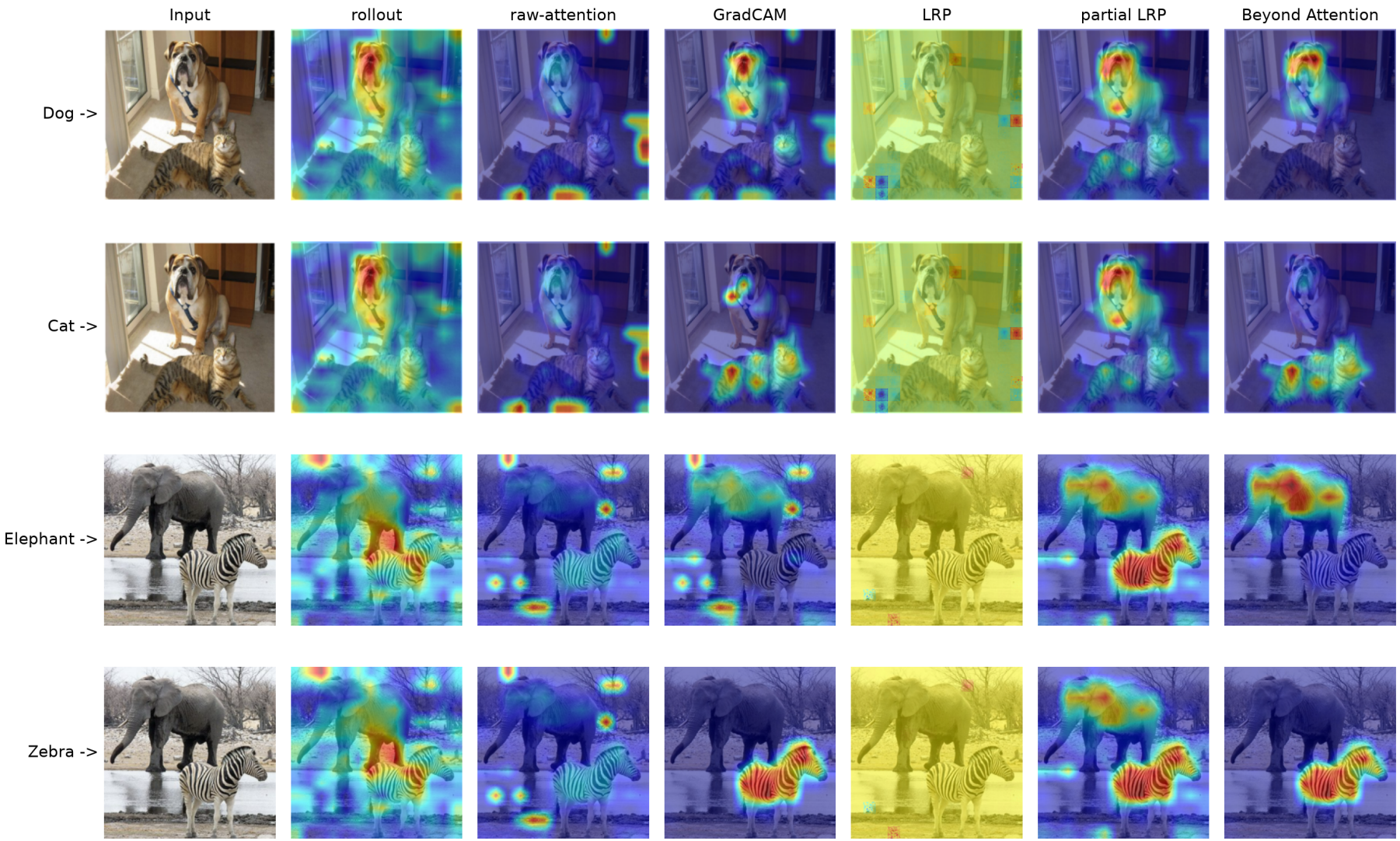


all layers, ∇ × relevance

class-specific

Left: ViT layers used. Right: result when we ask about **"cat"**.

Our reproduction of Fig. 3



Does it match the paper?

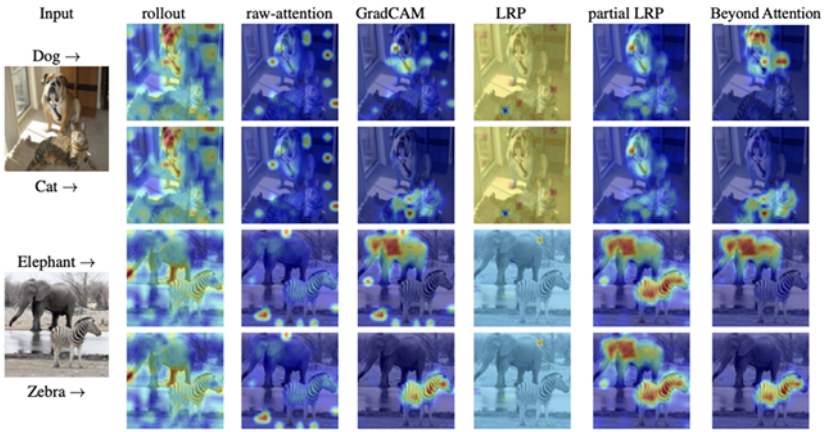
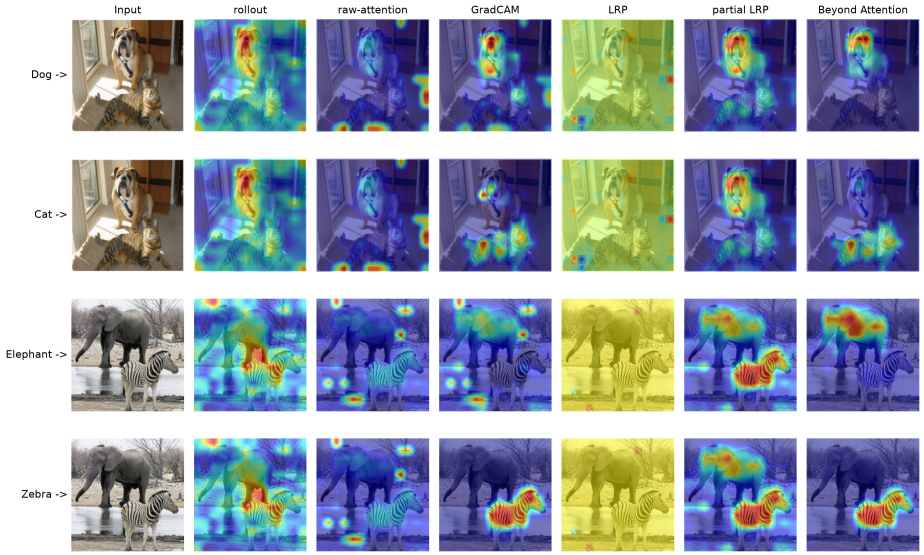


Figure 3: Class-specific visualizations of several attention-based methods, for each image results of two different class can bee seen Chefer et al. (2021a).

Paper (Fig. 3)



Ours

✓ rollout, raw-attention: same for both classes

✓ LRP: noisy

✓ **Beyond Attention**: cleanest, follows the class

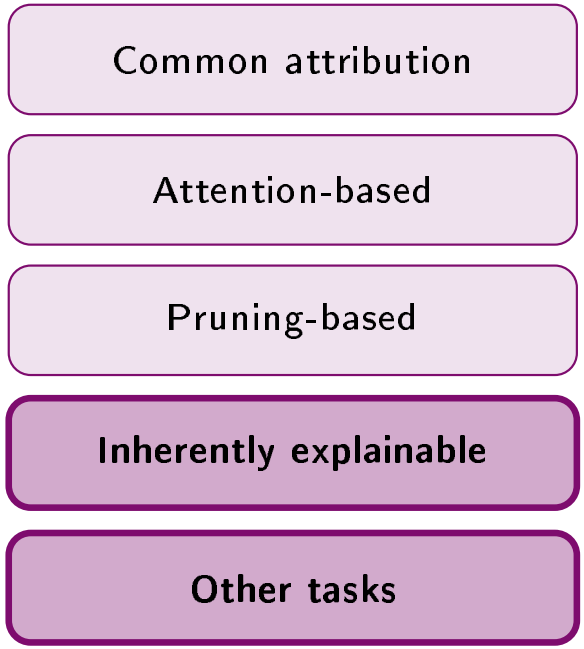
✓ same top-1: cat 52%, zebra 74%

≈ crop differs

Paper's claims reproduced (qualitatively)

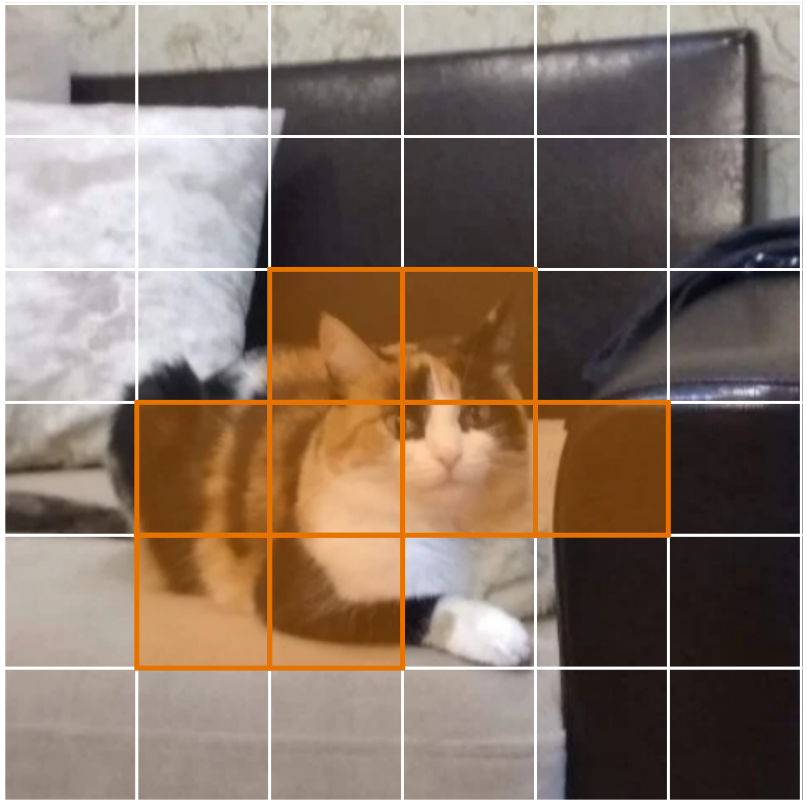
Outline of this part

3.4 Inherently explainable methods
3.5 Explainability of other tasks
4 Explanation evaluation
5–6 Tools and datasets
7–8 Discussion and conclusion

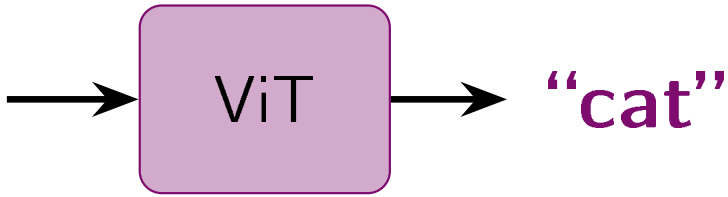


Taxonomy of the survey (Fig. 2): bold = this part

Running example: a cat on a sofa



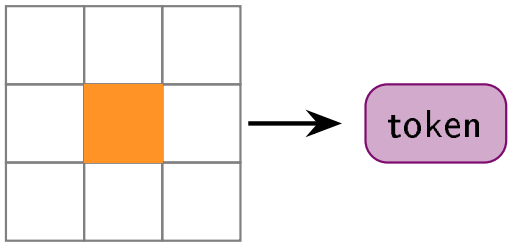
$6 \times 6 = 36$ patches: **8 cat** + 28 sofa



Which patches did the model really need to say “cat”?

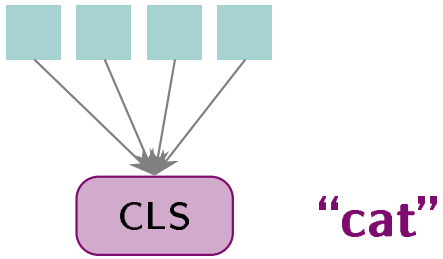
Photo: Kafun, CC BY-SA 4.0, Wikimedia Commons. A real ViT uses $14 \times 14 = 196$ patches; we draw 36.

Five key terms in pictures



Patch / token

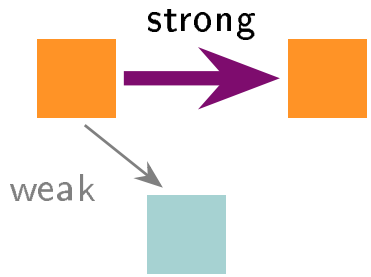
a piece of the image



Class token

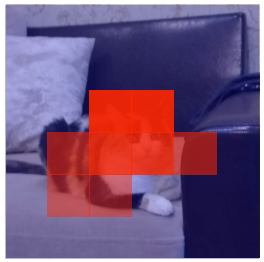
gathers all tokens, answers

“cat”



Attention

one token looks at another

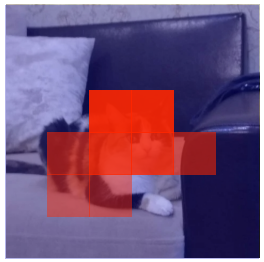


Saliency map

bright = important



model uses

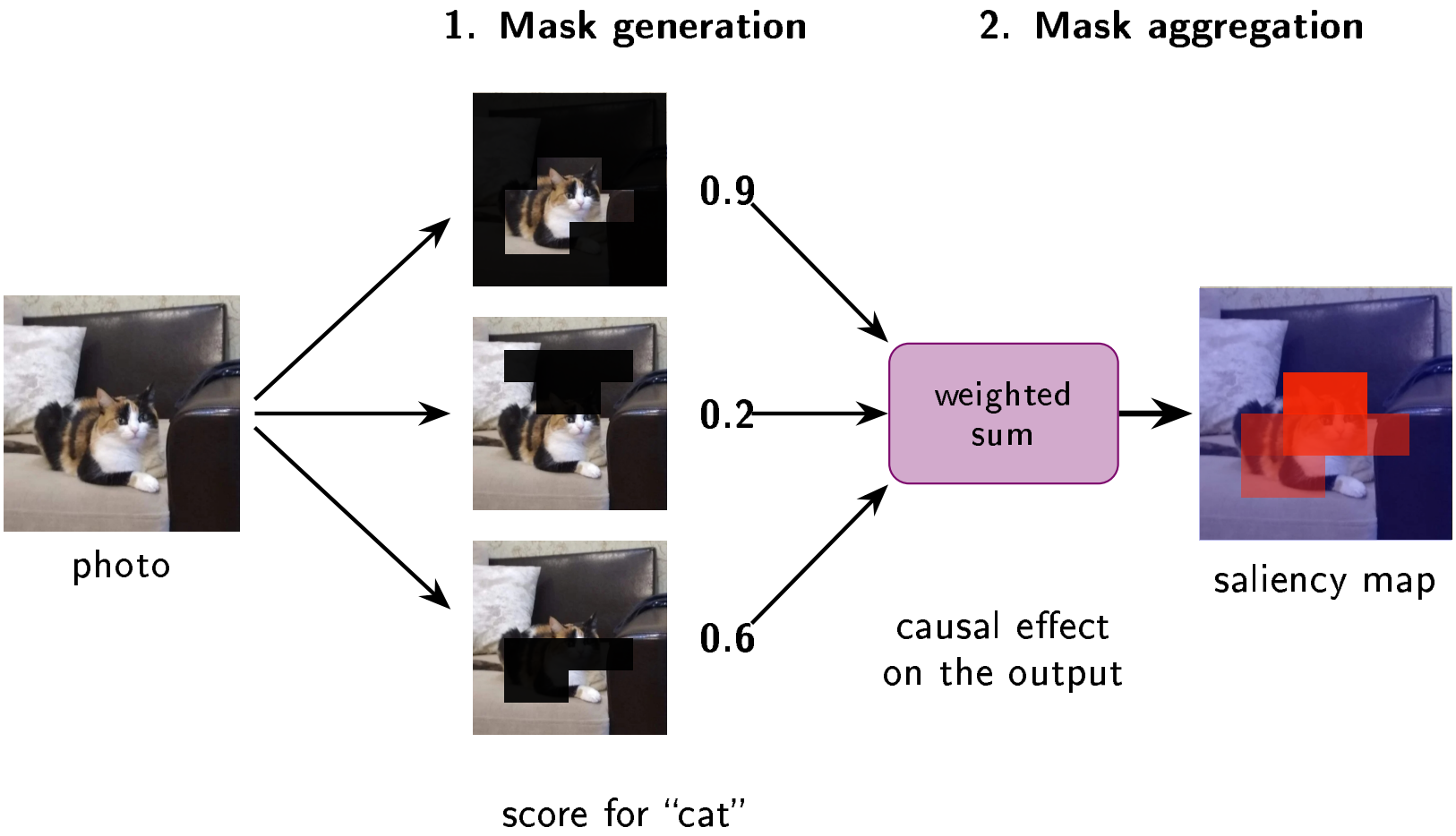


human expects

Faithful vs. plausible

3.4 Inherently explainable: ViT-CX

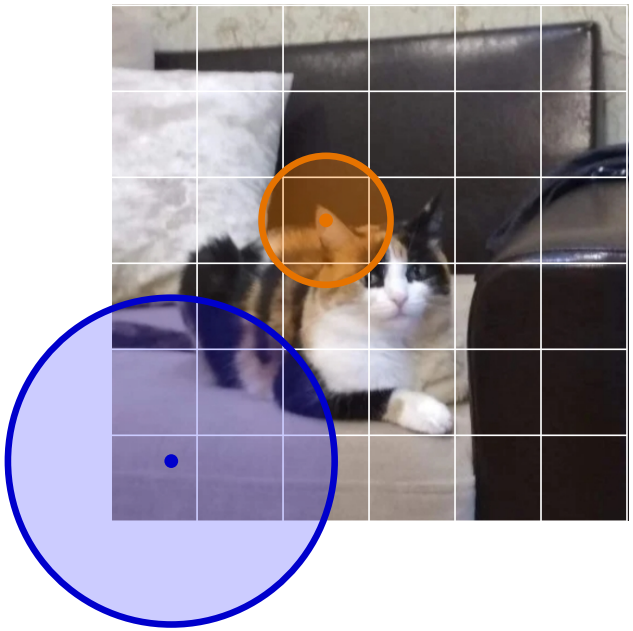
Self-explaining models: trees, rules, sparsity (less accurate)



Cat drawings: our own illustration, not from the paper.

Patch interactions & ViT-NeT

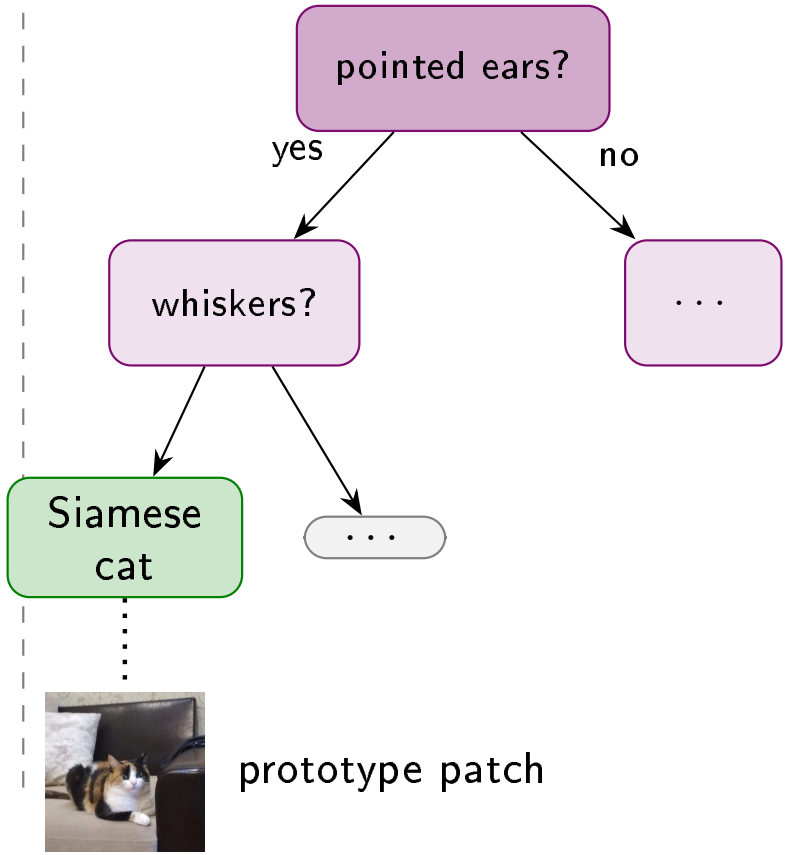
Patch interactions



object patch: small window

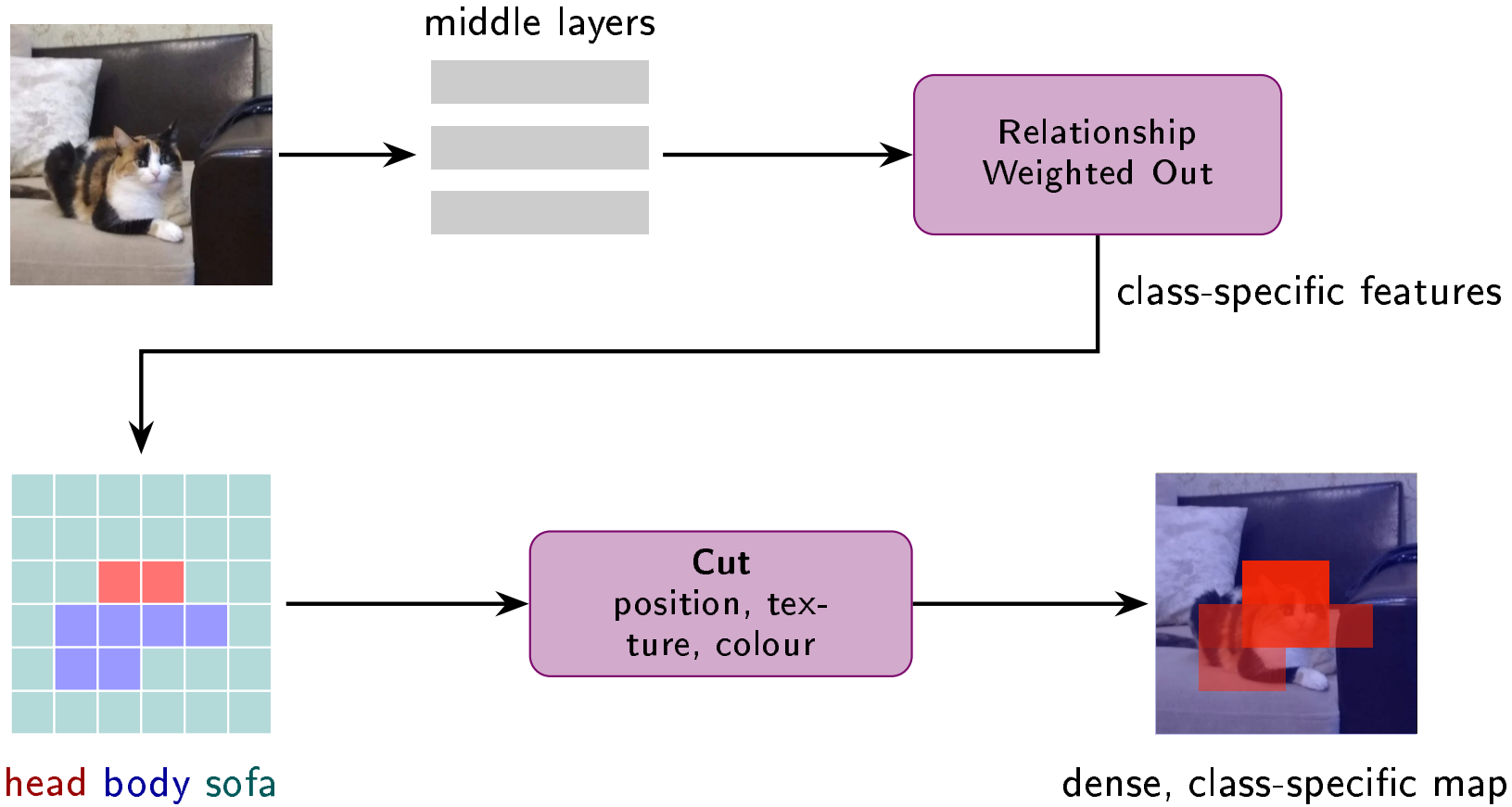
background patch: large window

ViT-NeT: tree decoder



Cat drawings: our own illustration, not from the paper.

R-Cut

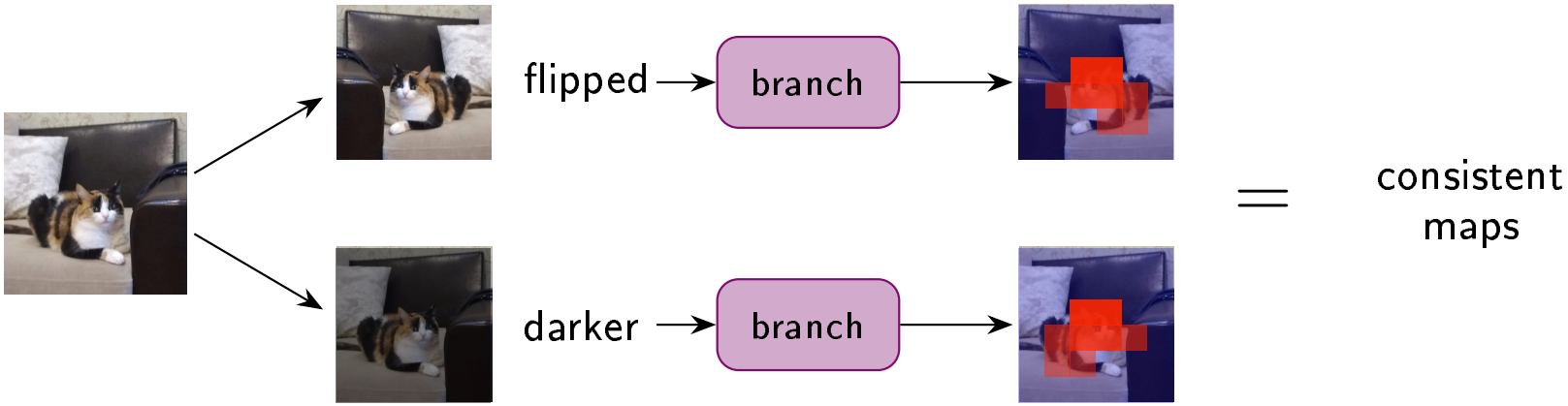


Evaluated with the **IoU test** (Section 4)

Cat drawings: our own illustration, not from the paper.

3.5 Other tasks: eX-ViT and DINO

eX-ViT: weakly supervised segmentation



DINO: no labels

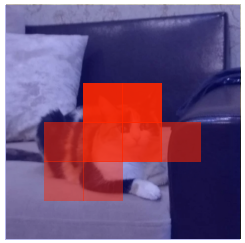


Cat drawings: our own illustration, not from the paper.

Generic Attention-model, ATMAN, Concept-Transformer

Generic

bi-modal, co-attention



patches **and** words

What is on the **sofa** ?

↓

"cat"

ATMAN

perturbs attention



weaken the face

score before

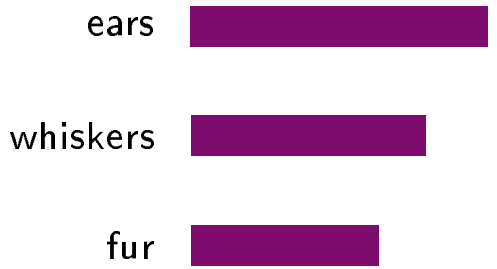


score after



Concept-Transformer

attention over concepts



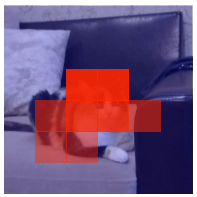
concepts, not pixels

↓

"cat"

Cat drawings: our own illustration, not from the paper.

4 Explanation evaluation

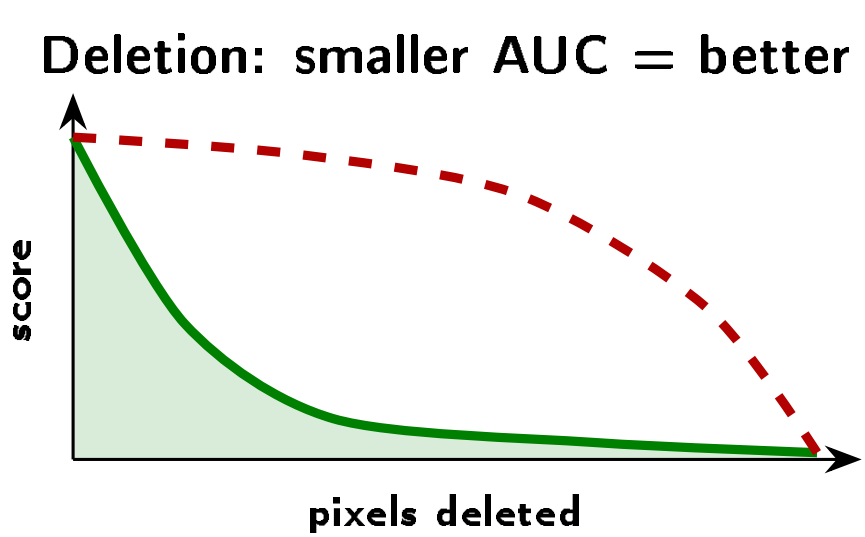


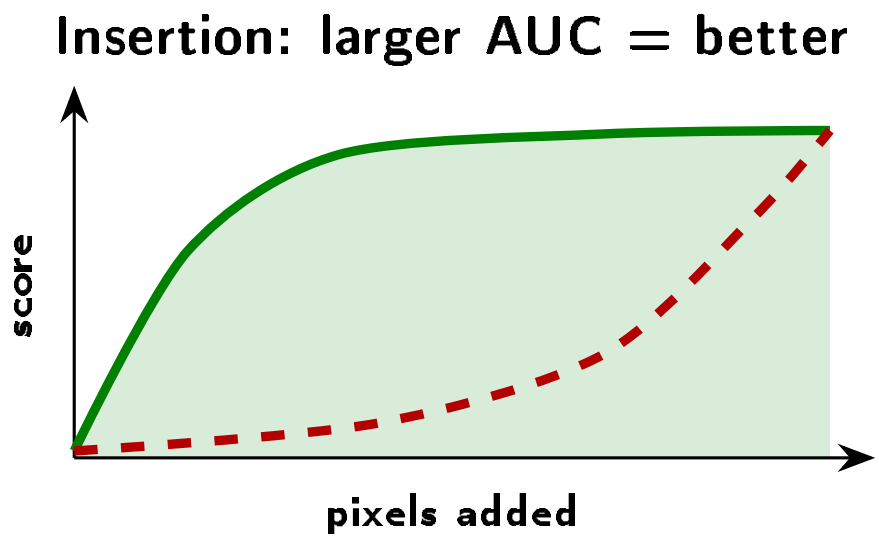
?

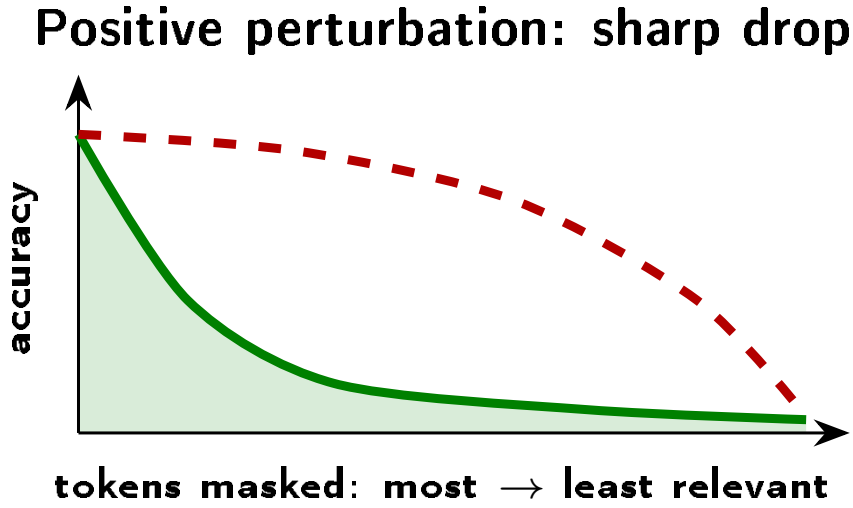
is this map right? → **Evaluative criteria**

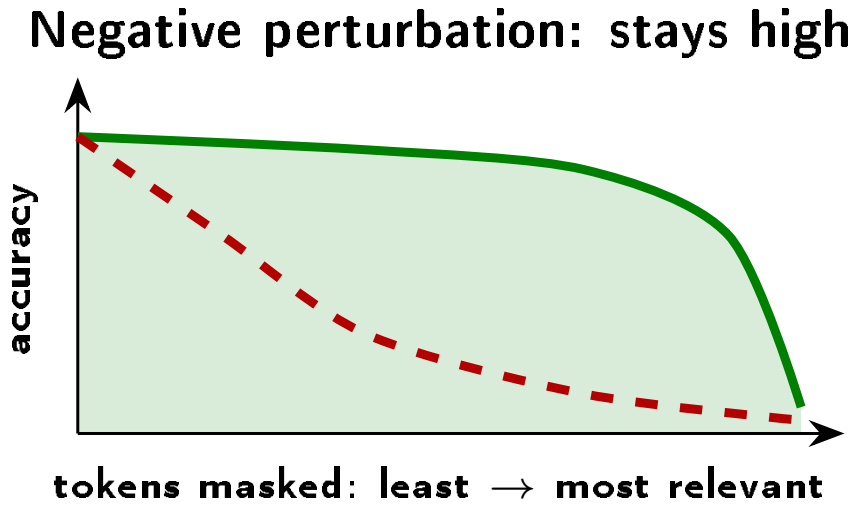
- Deletion / Insertion
- Effective Complexity
- Faithfulness
- (In)fidelity
- Sensitivity
- Sensitivity-n
- Perturbation tests
- IoU test
- (Grid) Pointing Game
- Segmentation tests

Deletion / Insertion and Perturbation tests





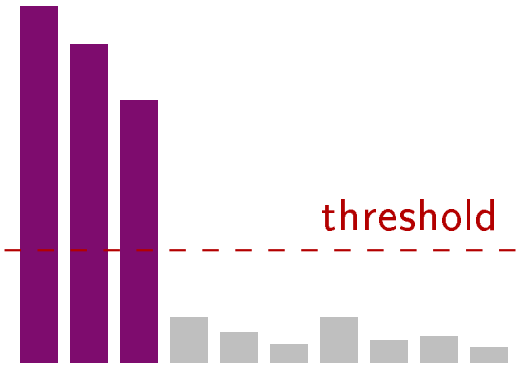




good map (cat first) bad map (sofa first)

Effective Complexity, Faithfulness, (In)fidelity

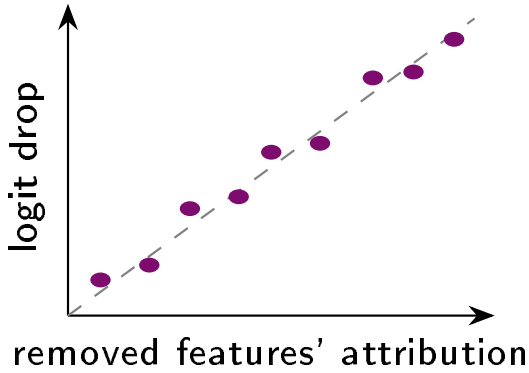
Effective Complexity



count values above it

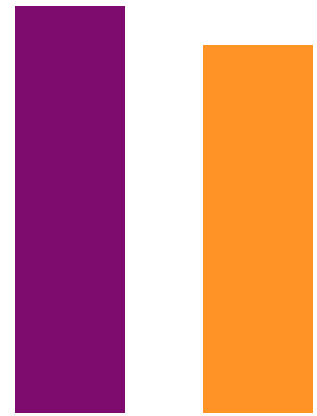
EC = 3: simple

Faithfulness



correlation in $[-1, 1]$

(In)fidelity



real change

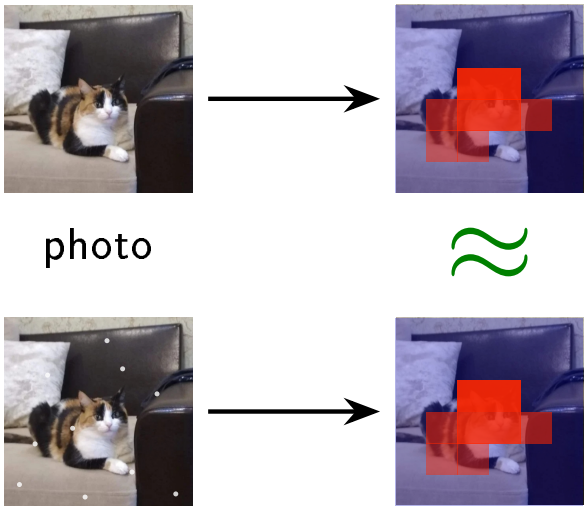
map predicts

close = low infidelity

Sensitivity and Sensitivity-n

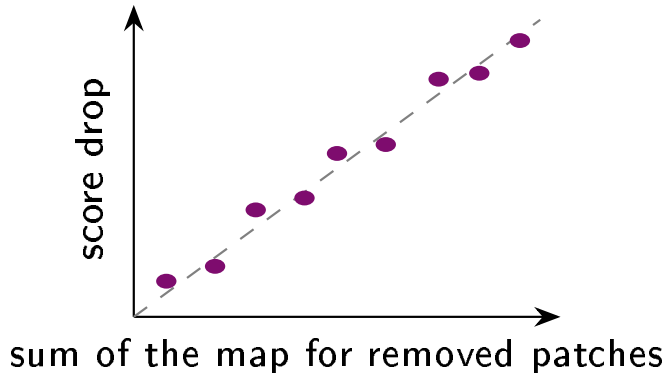
Sensitivity

tiny noise $\Rightarrow$ same map?



Sensitivity-n

remove n random patches

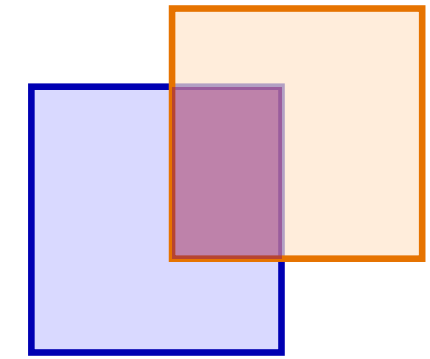


high correlation = accurate

Cat drawings: our own illustration, not from the paper.

IoU, Pointing Game and Segmentation tests

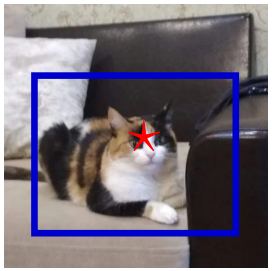
IoU test



true box map > 0.2

$$\text{IoU} = \frac{\cap}{\cup}$$

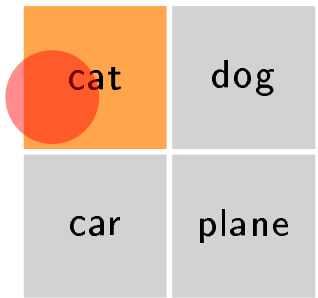
Pointing Game



brightest point in the box = hit

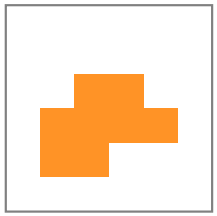
$$\frac{\#\text{hits}}{\#\text{hits} + \#\text{misses}}$$

Grid Pointing Game

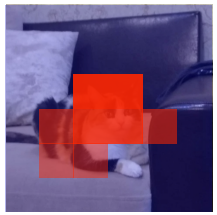


attribution should land on the cat cell

Segmentation tests



truth

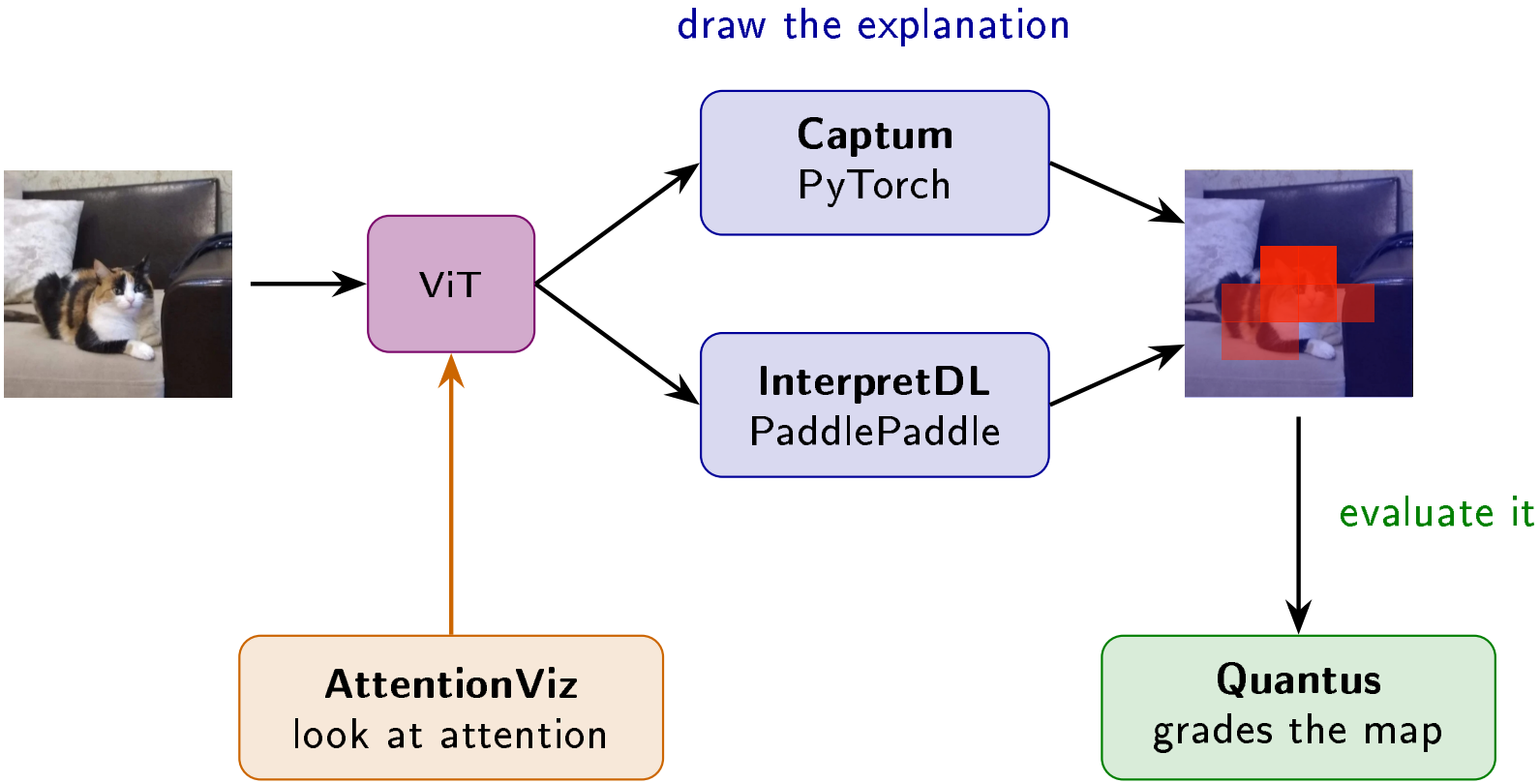


map

acc., mIoU, mAP, mF1

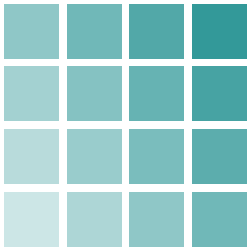
Cat drawings: our own illustration, not from the paper.

5 Tools and frameworks



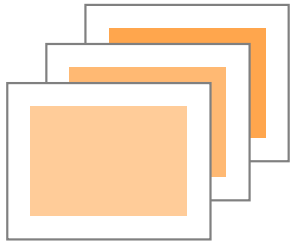
Cat drawings: our own illustration, not from the paper.

6 Datasets



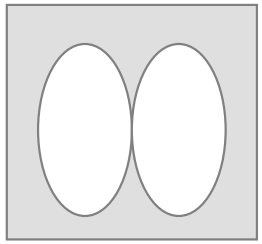
CIFAR-10

60,000 small images



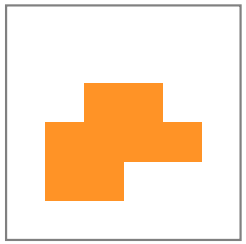
ImageNet

millions of images



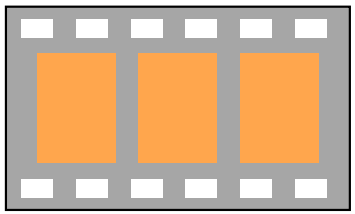
COVID-QU-Ex

33,920 chest X-rays



ImageNet-Segmentation

for segmentation tests

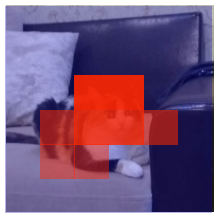


Kinetics-400

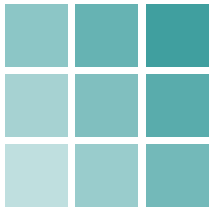
video classification

7 Discussion and future works

Attribution vs. visualization



local: which pixels?



global: what was learned?
less explored

Attention alone is not enough



raw attention

$\neq$ "cat"

Beyond classification

classi-fication

multi-modal

genera-tive

the last two: nascent

Use explanations for

debugging

improving

fairness

reliability

Cat drawings: our own illustration, not from the paper.

8 Conclusion

Inherent

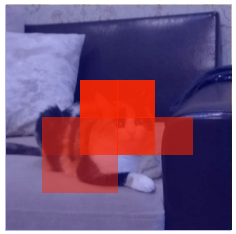
the model explains itself

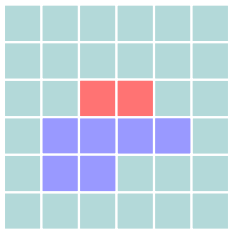
Other tasks

beyond classification: just starting

Evaluation

use several criteria





✓

Methods *find* the patches → criteria *check* them

Cat drawings: our own illustration, not from the paper.

References I


R. Kashefi, L. Barekatain, M. Sabokrou, F. Aghaeipoor. Explainability of vision transformers: A comprehensive review and new perspectives. *arXiv:2311.06786*, 2023.


H. Chefer, S. Gur, L. Wolf. Transformer interpretability beyond attention visualization. *CVPR*, 2021.


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Thank you!

Questions and discussion